



UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W24 24 x 9 (610 x 229)	1,830	76,200	70.4	2,930	9.55	24.3	1.87	4.75	154	2,520	15.7	257
	2,100	87,400	82.5	3,430	9.69	24.6	1.92	4.88	176	2,880	18.4	302
	2,370	98,600	94.4	3,930	9.79	24.9	1.95	4.95	196	3,210	20.9	342
	2,700	112,000	109.0	4,540	9.87	25.1	1.98	5.03	222	3,640	24.0	393
W24 24 x 12 (610 x 305)	2,987.3	124,400	203.5	8,471	10.08	25.6	2.63	6.68	248.9	4,079.5	33.9	555.6
	3,635.3	151,300	254.0	10,570	10.15	25.78	2.68	6.81	299.1	4,902.2	42.0	688.4
	4,986.3	207,571	380.5	15,838	10.28	26.1	2.84	7.22	400.2	6,559	62.0	1,017
W24 24 x 12 ³ / ₄ (610 x 324)	3,100	129,000	259	10,800	10.1	25.7	2.91	7.39	258	4,230	40.7	667
	3,540	147,000	297	12,400	10.1	25.7	2.94	7.47	291	4,770	46.5	762
	4,020	167,000	340	14,200	10.2	25.9	2.97	7.54	329	5,390	53.0	869
	4,580	191,000	391	16,300	10.3	26.2	3.01	7.65	371	6,080	60.5	991
	5,170	215,000	443	18,400	10.4	26.4	3.05	7.75	414	6,780	68.4	1,120
W27 27 x 10 (686 x 254)	2,850	119,000	106	4,410	10.7	27.2	2.07	5.26	213	3,490	21.2	347
	3,270	136,000	124	5,160	10.9	27.7	2.12	5.38	243	3,980	24.8	406
	3,620	151,000	139	5,790	11.0	27.9	2.15	5.46	267	4,380	27.8	456
	4,090	170,000	159	6,620	11.0	27.9	2.18	5.54	299	4,900	31.5	516
W27 27 x 14 (686 x 356)	5,630	234,000	443	18,400	11.4	29.0	3.21	8.15	411	6,740	63.5	1,040
	6,280	261,000	497	20,700	11.5	29.2	3.24	8.23	455	7,460	70.9	1,160
	6,990	291,000	555	23,100	11.6	29.5	3.26	8.28	502	8,230	78.8	1,290
W30 30 x 10 ¹ / ₂ (762 x 267)	4,000	166,000	128	5,330	11.7	29.7	2.10	5.33	269	4,410	24.5	401
	4,470	186,000	146	6,080	11.9	30.2	2.15	5.46	299	4,900	27.9	457
	4,930	205,000	164	6,830	12.0	30.5	2.19	5.56	329	5,390	31.3	513
	5,360	223,000	181	7,530	12.1	30.7	2.23	5.66	355	5,820	34.4	564
	5,770	240,000	196	8,160	12.2	31.0	2.25	5.71	380	6,230	37.2	610
W30 30 x 15 (762 x 381)	8,200	341,000	598	24,900	12.7	32.3	3.43	8.71	539	8,830	79.8	1,310
	9,170	382,000	673	28,000	12.8	32.5	3.46	8.79	598	9,800	89.5	1,470
	10,300	429,000	757	31,500	12.9	32.8	3.49	8.86	663	10,900	100	1,640
W33 33 x 11 ¹ / ₂ (838 x 292)	5,900	246,000	187	7,780	13.0	33.0	2.32	5.89	359	5,880	32.6	534
	6,710	279,000	218	9,070	13.2	33.5	2.38	6.05	406	6,650	37.9	621
	7,450	310,000	246	10,200	13.4	34.0	2.43	6.17	448	7,340	42.7	700
	8,160	340,000	273	11,400	13.5	34.3	2.47	6.27	487	7,980	47.2	773
W33 33 x 15 ³ / ₄ (838 x 400)	11,500	479,000	749	31,200	14.0	35.6	3.56	9.04	684	11,200	95.2	1,560
	12,800	533,000	840	35,000	14.1	35.8	3.59	9.12	757	12,400	106	1,740
	14,200	591,000	932	38,800	14.1	35.8	3.63	9.22	829	13,600	118	1,930
W36 36 x 12 (914 x 305)	7,800	325,000	225	9,370	14.0	35.6	2.38	6.05	439	7,190	37.7	618
	9,040	376,000	270	11,200	14.3	36.3	2.47	6.27	504	8,260	45.1	739
	9,760	406,000	295	12,300	14.4	36.6	2.50	6.35	542	8,880	49.1	805
	10,500	437,000	320	13,300	14.5	36.8	2.53	6.43	580	9,500	53.2	872
	11,300	470,000	347	14,400	14.5	36.8	2.55	6.48	623	10,200	57.6	944
	12,100	504,000	375	15,600	14.6	37.1	2.56	6.50	664	10,900	61.9	1,010
	13,200	549,000	411	17,100	14.6	37.1	2.58	6.55	719	11,800	67.5	1,110
W36 36 x 16 ¹ / ₂ (914 x 419)	15,000	624,000	940	39,100	14.9	37.8	3.73	9.47	837	13,700	114	1,870
	16,100	670,000	1,010	42,000	15.0	38.1	3.75	9.52	895	14,700	123	2,020
	17,300	720,000	1,090	45,400	15.0	38.1	3.78	9.60	953	15,600	132	2,160
	18,900	787,000	1,200	49,900	15.1	38.4	3.81	9.68	1,030	16,900	144	2,360
	20,300	845,000	1,300	54,100	15.2	38.6	3.83	9.73	1,110	18,200	156	2,560